
MHF 4U1

CHAPTER 7: EXPONENTIAL AND LOGARITHMIC FUNCTIONS

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CHAPTER 7 TEST DATE:

7.1 EQUIVALENT FORMS OF EXPONENTIAL FUNCTIONS

- Exponential functions and expressions can be expressed in different ways by changing the base.
- Changing the base of one or more exponential expressions is a useful technique for solving exponential equations

EXAMPLES:

1. Express in terms of a power with a base of 3.
- a) 27 b) 9^4

2. Express in terms of a power with a base of 2.
- a) $\sqrt{32} \times (\sqrt[3]{8})^5$ b) 18

This cannot be solved by inspection as in previous examples. **To express 18 as a power of 2, solve the equation $2^k = 18$ for k.**

3. Solve each equation without using logarithms.
- a) $3^{x+4} = 81^x$ b) $8^{4x-1} = 32^{x+2}$

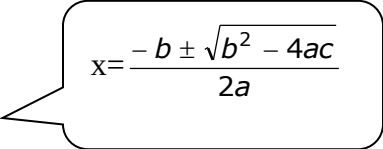
c) $64^{m-3} = (\sqrt{128})^{2m}$

7.2 TECHNIQUES FOR SOLVING LOGARITHMIC EQUATIONS

- An equation maintains balance when the common logarithm is applied to both sides
- The **power law of logarithms** is a useful tool for solving a variable that appears as part of the exponent.
- When a quadratic equation is obtained, methods such as *factoring* and applying the *quadratic formula* may be useful.
- Some algebraic methods of solving exponential eq'ns leads to *extraneous roots*, which are not valid solutions to the original equations.

EXAMPLES: Part 1

1. Solve the equation $10 = \left(\frac{1}{4}\right)^{3m}$. Express solution in both exact and approximate form, to two decimal places. Check for extraneous roots.


$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

2. Solve exactly: $2^{m-2} = 3^{m+1}$.

3. Solve & check for extraneous roots: $4^{2x} - 4^x = 20$

Try it now!**Let's review the key properties of logarithms!**

1. The logarithmic function $y = \log_b(x)$ is the inverse of the _____ function _____, where b _____.
2. The statement $\log_b(m) = n$ is equivalent to _____
3. The logarithm, $\log_b(x)$ is defined only when _____
4. The following statements hold true for logarithms with $b > 0$ and $b \neq 1$:
 - $b^{\log_b(n)} = \underline{\hspace{2cm}}$, $n \underline{\hspace{2cm}}$
 - $\log_b(b^n) = \underline{\hspace{2cm}}$
 - $\log_b(1) = \underline{\hspace{2cm}}$
 - $\log_b(m) = \log_b(n)$ if and only if _____ and _____

5. Solve.

a) $5^{\log_5\left(\frac{1}{25}\right)}$

b) $8^{\log_4(x+1)} = 27$

7.3 THE PRODUCT AND QUOTIENT LAWS OF LOGARITHMS

Recall: The Exponent Laws:



These laws can be extended to logarithms. Laws of Logarithms can be used to simplify expressions and solve equations.

THE PRODUCT LAW of LOGARITHMS:

$$\log_b x + \log_b y = \log_b(xy) \text{ for } b > 0, b \neq 1, x > 0, y > 0$$

PROOF:

Let $\log_a m = x$ and Let $\log_a n = y$

Then $a^x = m$ and $a^y = n$

$$\begin{aligned} \text{Therefore, } \log_a(mn) &= \log_a(a^x \cdot a^y) \\ &= \log_a(a^{x+y}) \\ &= x + y \\ &= \log_a m + \log_a n \end{aligned}$$

THE QUOTIENT LAW of LOGARITHMS:

$$\log_b x - \log_b y = \log_b\left(\frac{x}{y}\right) \text{ for } b > 0, b \neq 1, x > 0, y > 0$$

PROOF:

Let $\log_a m = x$ and Let $\log_a n = y$

Then $a^x = m$ and $a^y = n$

$$\begin{aligned} \text{Therefore, } \log_a\left(\frac{m}{n}\right) &= \log_a\left(\frac{a^x}{a^y}\right) \\ &= \log_a(a^{x-y}) \\ &= x - y \\ &= \log_a m - \log_a n \end{aligned}$$

Examples:

1. Simplify using the laws of logarithms.

a) $\log_8 16 + \log_8 4$

b) $\log_3 54 - \log_3 2$

2. Rewrite into a single logarithm.

$$6\log_a x + 7\log_a y$$

3. Express as a sum/difference of logarithms.

$$\log_a \left(\frac{q^5 r^8}{s^2} \right)$$

4. Simplify. State any restrictions on the variables:

a) $\log_{10} x + \frac{1}{2}\log_{10} y - 3\log z$

b) $\log(x^2 - 25) - \log(x^2 + 3x - 10)$

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Try it now!

Solve exactly: $2(5^{6x}) - 9(5^{4x}) + 10(5^{2x}) - 3 = 0$

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Try it now!

If $\left(\frac{1}{4}\right)^{x-y} = 64$ and $\log_{x+y} 10 = 3$, find the values of x and y . Round answers to 2 dec. places.

7.4 TECHNIQUES FOR SOLVING LOGARITHMS

$$\text{Recall: } y = \log_b x \longleftrightarrow x = b^y$$

- It is possible to solve an equation involving logarithms by expressing **both sides** as a **logarithm of the same base**, or **vice versa**:

$$\text{If } a = b, \text{ then } \log a = \log b,$$

$$\text{If } \log a = \log b, \text{ then } a = b$$

- When a quadratic equation is obtained from solving a logarithm, methods such as *factoring* and applying the *quadratic formula* may be useful. However keep in mind that some algebraic methods of solving exponential eq'ns leads to *extraneous roots*, which are not valid solutions to the original equations.
- Always check your final answers to confirm validity!**

Examples:

- Find the roots of the equation.

a) $\log(x + 3) = 1$

b) $\log_4(3x - 1) = 3$

c) $\log_3 x - \log_3 5 = \log_3 6$

- Solve. Identify and reject any extraneous roots.

a) $\log_2(x - 2) + \log_2(x + 1) = 2$

b) $\log_3 x + \log_3(x - 1) = \log_3(2x)$

c) $\log_{10}(1 + \sqrt{x}) = 1 + \log_{10}(1 - \sqrt{x})$

7.5 MATHEMATICAL MODELS WITH EXPONENTIAL AND LOGARITHM FUNCTIONS

- Exponential and logarithmic equations often appear in contexts that involve continuous growth or decay.

1. POPULATION GROWTH: ($y = ab^x$)

$$P_L = P_0 b^{\frac{t}{d}}$$

where P_L = population remaining after t years ,
 P_0 = original/ initial population,
 t = the elapsed time
 d = doubling time (if $b = 2$)
 $b = 2$ if doubling; 3 if tripling,...

t & d must
be in the
same units

Examples:

- The population in a small town in northern Ontario has a population of 10 000. After 40 years, the population is 30 000. What is the doubling period?

2. FINANCE:

$$A = P(1 + i)^n$$

where A = amount remaining (\$)

P = principal (original) amount (\$)

i = interest rate, per compounding period

n = compounding period

Note: yearly(1X /year), semi-annually (2 X/year), quarterly (4 X/year), monthly (12 X/year)...

Example:

- Find the length of time required for an investment of \$1000 to amount to \$1500 at a rate of 9% per year compounded quarterly.

3. HALF-LIFE:

$$A(t) = A_0 \left(\frac{1}{2} \right)^{\frac{t}{h}}$$

where: A(t)= new amount
A₀ = original amt
h = half-life time
t = the elapsed time
h & t must be in same units

$$A(t) = A_0 \left(\frac{1}{2} \right)^{\frac{t}{h}}$$

where: A(t)= new amount

where: $A(t)$ = new amount

A_0 = original amt

h = half-life time

t = the elapsed time

h & t must be in same units

Example:

3. A 50 mg sample of radioactive iodine decays to 23 mg after 12 minutes. Determine the half-life of the iodine.

4. Polonium-218 has a half-life of 3.1 minutes. How long will it take for this sample to decay to 25% of its initial amount of 120 mg?

3. A 50 mg sample of radioactive iodine decays to 23 mg after 12 minutes. Determine the half-life of the iodine.

4. Polonium-218 has a half-life of 3.1 minutes. How long will it take for this sample to decay to 25% of its initial amount of 120 mg?

Try it now!

CHAPTER 7 REVIEW QUESTIONS: TRUE AND FALSE

- For any values $a > 0$ and $b > 0$, it can be shown that $a = b^{\frac{\log a}{\log b}}$.
- The solution to the equation $3^{x+4} = 16$ is $x = \frac{\log 16 - 4 \log 3}{\log 3}$.
- $\log_a xy = (\log_a x)(\log_a y)$
- $\log_a \frac{x}{y} = \frac{\log_a x}{\log_a y}$
- $\log x \pm \log y = \log(x \pm y)$
- $\left(\log_3 17\right)^4 = 4 \log_3 17$
- $\log_a \frac{1}{x} = -\log_a x$

Try it now!

1. Express the following logarithm in terms of $\log_2(x)$, $\log_2(y)$, and/or $\log_2(z)$.

a) $\log_2(2xy^3)$

$$\text{b) } \log_2\left(\frac{4xy^3}{z}\right)$$

2. Use the laws of logarithms to simplify and evaluate the following logarithmic expression (without a calculator).

a) $\log_2(3) - \log_2(9) + \log_2(6)$

b) $\log_2(12) + \log_2(3) - \log_2(9)$